

SEMARANG MARITIME, Indonesia

WMO: 968370

Lat: 6.9508S

Lon: 110.4183E

Elev: 3

StdP: 101.29

Time Zone: 7.00 (E07)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	23.6	24.1	16.0	11.3	30.0	17.2	12.3	29.1	10.5	27.2	9.9	27.3	1.6	110	0.380

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	7.0	34.2	24.5	33.6	24.7	33.1	24.9	27.3	31.2	27.1	31.1	26.9	30.9	3.7	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.6	30.1	25.9	21.2	29.9	25.7	20.9	29.7	86.7	31.5	85.4	31.3	84.4	31.1	33.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	6.7	5.7	DB	21.9	35.9	1.1	1.4	21.1	36.9	20.5	37.8	19.9	38.6	19.1	39.6	(4)
(5)			WB	18.8	29.3	1.3	1.4	17.8	30.3	17.1	31.1	16.3	31.9	15.4	33.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	28.6	27.6	27.6	28.2	29.0	29.2	28.7	28.4	28.6	29.2	29.6	28.9	28.2	(6)
(7)		DBStd	1.06	0.94	0.93	0.78	0.80	0.84	0.80	0.76	0.73	0.86	1.04	1.13	0.92	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6797	547	492	566	569	595	562	571	578	577	607	568	566	(10)
(11)		CDD18.3	3756	288	259	307	319	337	312	313	319	327	349	318	307	(11)
(12)		CDH23.3	43761	3036	2713	3501	3831	4169	3762	3632	3720	3886	4317	3755	3439	(12)
(13)		CDH26.7	16946	860	750	1187	1525	1793	1557	1452	1528	1676	1941	1515	1161	(13)
(14)	Wind	WSAvg	2.2	2.8	2.7	2.1	1.8	2.1	2.0	2.2	2.5	2.3	2.1	1.8	2.0	(14)
(15)	Precipitation	PrecAvg	2162	383	314	227	178	174	83	80	48	86	145	203	249	(15)
(16)		PrecMax	3507	894	556	501	399	378	256	329	191	288	396	425	537	(16)
(17)		PrecMin	1494	78	106	29	51	14	1	0	1	0	0	57	84	(17)
(18)		PrecStd	613	217	139	106	84	105	75	80	55	88	117	85	112	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.4	32.2	32.3	33.5	33.7	33.7	33.5	33.7	35.0	35.4	34.9	33.1	(19)
(20)			MCWB	26.6	26.6	26.5	26.2	25.7	25.0	23.9	23.4	23.2	24.1	25.4	26.4	(20)
(21)		2%	DB	31.3	31.2	31.6	32.8	33.2	33.0	32.8	33.1	34.1	34.2	33.6	32.1	(21)
(22)			MCWB	26.3	26.3	26.4	26.3	25.7	24.9	23.8	23.6	23.4	24.6	25.7	26.4	(22)
(23)		5%	DB	30.5	30.5	31.1	32.1	32.6	32.4	32.3	32.6	33.2	33.3	32.6	31.3	(23)
(24)			MCWB	26.1	26.1	26.3	26.3	25.8	25.0	24.0	23.8	23.9	25.1	26.1	26.3	(24)
(25)		10%	DB	29.9	29.9	30.6	31.5	32.1	31.8	31.7	31.9	32.4	32.5	31.7	30.7	(25)
(26)			MCWB	25.9	25.9	26.2	26.3	25.8	25.1	24.2	24.1	24.3	25.4	26.1	26.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.2	27.1	27.4	27.7	27.7	27.2	26.7	27.1	27.0	27.1	27.6	27.2	(27)
(28)			MCD B	30.8	30.9	30.9	31.2	31.7	31.1	30.6	31.2	31.2	31.6	31.5	30.9	(28)
(29)		2%	WB	26.7	26.6	27.0	27.2	27.2	26.7	26.2	26.2	26.3	26.7	27.0	26.8	(29)
(30)			MCD B	30.4	30.2	30.7	31.0	31.4	30.8	30.3	30.4	30.7	31.3	31.5	30.8	(30)
(31)		5%	WB	26.3	26.3	26.7	26.9	26.8	26.5	25.9	25.7	26.0	26.4	26.6	26.6	(31)
(32)			MCD B	29.8	29.7	30.3	30.8	31.0	30.6	30.0	30.0	30.4	31.1	31.0	30.4	(32)
(33)		10%	WB	26.1	26.1	26.4	26.6	26.5	26.1	25.5	25.4	25.6	26.1	26.3	26.2	(33)
(34)			MCD B	29.4	29.4	29.9	30.4	30.6	30.3	29.6	29.7	30.1	30.8	30.7	29.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.6	4.5	4.9	5.6	6.2	6.7	7.2	7.7	7.8	7.0	6.2	5.1	(35)
(36)			MCDBR	5.4	5.3	5.5	6.3	6.7	7.5	7.9	8.2	8.8	8.0	7.4	5.9	(36)
(37)			MCWBR	2.4	2.4	2.4	2.4	2.5	3.0	3.4	3.7	3.7	3.1	2.9	2.5	(37)
(38)		5% WB	MCDBR	5.0	4.9	5.1	5.7	6.1	6.2	6.6	7.2	7.5	7.0	6.4	5.4	(38)
(39)			MCWBR	2.4	2.4	2.5	2.5	2.6	2.7	3.0	3.7	3.4	3.1	2.9	2.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.533	0.539	0.539	0.549	0.522	0.511	0.514	0.545	0.576	0.656	0.639	0.575	(40)	
(41)		taud	2.099	2.084	2.085	2.041	2.107	2.128	2.106	2.002	1.925	1.741	1.792	1.963	(41)	
(42)		Ebn at Noon	820	814	802	766	757	751	757	758	761	715	729	780	(42)	
(43)		Edh at Noon	171	174	171	171	153	146	152	176	197	242	230	194	(43)	
(44)	All-Sky Solar Radiation	RadAvg	4.57	4.63	5.02	5.13	5.06	4.87	5.18	5.79	6.20	5.91	5.22	4.60	(44)	
(45)		RadStd	0.46	0.40	0.30	0.31	0.28	0.33	0.31	0.22	0.34	0.52	0.49	0.39	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		+0.32	N/A	N/A	+0.39	+0.35	+0.47	N/A	N/A	N/A	N/A

Nomenclature: See separate page